

Plan of Work

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Part A. BEP Research Proposal (2-6 pages)

Project Title

AI-supported operational purchasing at Hytch-Pommec

1. Introduction

Hytech-Pommec develops and manufactures hyperbaric oxygen and life support systems. Purchasing ensures that the company receives the right materials from suitable suppliers at the required time, at the required price, and of the required quality. Purchasing generally encompasses strategic, tactical, and operational activities. Strategic purchasing concerns long-term purchasing policies and their alignment with organisational objectives. Tactical purchasing covers specification, supplier selection and contracting, while operational purchasing covers ordering, monitoring and evaluation (Bäckstrand et al., 2019). These tasks require manual information handling, verification, and judgement, but their relative workload contributions and underlying causes have not yet been established. Hytech-Pommec would like to apply AI tools to operational purchasing in order to reduce the operational buyer's workload, especially the workload associated with the operational buyer's manual purchasing tasks, checking, and information handling.

This project focuses primarily on operational purchasing and aims to investigate how AI-supported tools can reduce the workload of operational buyers while maintaining purchasing quality. Strategic purchasing will not be optimized by the AI-supported solution, in line with the company's requirements. Relevant tactical purchasing activities will be examined, where necessary, to understand information flows, decision-making responsibilities, and the division of work within the purchasing department. For instance, some of the information and decisions used in operational purchasing originate in tactical purchasing. The tactical buyer's relevant work and its connections with the operational buyer's work therefore form part of the investigation. The project will study the information exchanges between them and their distinct decision responsibilities, and examine how those connections help explain observed handling time or repeated work. Changes to the tactical buyer's work will be considered where the improvement affects their tasks or responsibilities. Nevertheless, the study of the tactical buyer's purchasing process aims to help understand and reduce the operational buyer's workload.

A reduction in workload will count as an improvement only if the outcome of the selected purchasing activity continues to meet its relevant requirements, with quality evaluated against current practice. Depending on the selected activity, relevant requirements may include correct items and quantities in purchase orders, accurate identification of discrepancies between supplier confirmations and the corresponding orders, or complete and correct drawing sets for customised articles. Once the activity has been selected, we will agree on its specific quality criteria with the relevant buyers before

developing and testing the solution. A single purchasing task or a coherent process step will be selected for detailed redesign and optimization using AI. Depending on the selected purchasing task, our proposed solution could support tasks such as extracting and validating information, comparing purchasing documents, identifying missing or inconsistent data, or assisting employees in making routine decisions.

Following Bowling and Kirkendall (2012), this project will distinguish between the amount of work, a quantitative measure of workload, and the difficulty of work, a qualitative measure of workload. For the amount of work, indicators may include task frequency, case volume, processing time, and rework. For the difficulty of work, the buyer's evidence about judgement and problem solving will be considered. Processing time provides information about the time required to perform the work, but does not by itself represent the full workload construct. Purchasing quality refers to the extent to which the outcome of a purchasing activity fulfills the requirements relevant to that activity. The specific workload indicators and quality criteria for evaluation will be justified after selecting the improvement component and specified before development and formal testing. Van Weele's purchasing-process model, discussed by Bäckstrand et al. (2019), locates tasks within specification, supplier selection, contracting, ordering, monitoring, and evaluation. It provides a framework for examining operational purchasing, relevant tactical purchasing tasks, and the connections between them. The workload indicators will be examined within these process stages, which classify purchasing activities rather than measure workload themselves.

2. Research question

Main research question

To what extent can an AI-supported solution reduce the operational buyer's workload at Hytech-Pommec without reducing the quality of the purchasing outcome?

Sub research questions (if applicable)

1. What is the standard workflow of the purchasing process at Hytech-Pommec, and what is the relationship between operational and tactical purchasing tasks?
2. What conditions and control measures are required for the proposed solution to be implemented reliably in the purchasing workflow, including relevant operational and tactical purchasing tasks?
3. Which parts of the purchasing workflow contribute most to the operational buyer's workload in terms of proposed measures?
4. Which of these activities offers the greatest potential for AI-supported improvement, considering workload contribution, business relevance, technical feasibility, and the need for human expertise?
5. To what extent does the proposed AI-supported solution reduce workload while maintaining the required quality of the purchasing activity?

3. Empirical context (incl. company description)

Hytech-Pommec develops and manufactures hyperbaric oxygen and life support systems. Purchasing ensures that the company receives the right materials from suitable suppliers at the required time, price, and quality. Strategic and tactical purchasing includes supplier selection, contracts, sourcing decisions, and supplier relationships. Operational purchasing executes purchasing decisions through requests, orders, monitoring, and information checks. *Exact* supports order administration, allocation to underlying demand, and supplier information.

This BEP will be conducted under the supervision of Johan, the company supervisor and purchasing manager. Johan will provide the business context of Hytech-Pommec's purchasing requests and offer support for access and feasibility decisions.

Hytech-Pommec has one operational buyer, named Arno, and one tactical buyer, named Dennis. Although, in theory, operational and tactical purchasing are distinct and complementary, this is not always the case in a company's day-to-day operations. The manager of the purchasing department has identified overlap between Arno's and Dennis's work, including coverage during absences. Therefore, job titles, i.e., operational buyer and tactical buyer, alone do not accurately reflect the actual tasks and work content at Hytech-Pommec. Instead, task purpose and observed responsibilities will determine process-stage assignments.

To develop a solution for Hytech-Pommec's operations, the study will observe Arno's operational work and Dennis's relevant tactical work to document the actual purchasing workflow. In this study, the operational purchasing work is defined by the flow that runs from the purchasing request to supplier confirmation, recorded as Bevestigd in Exact (the business software Hytech-Pommec uses). Relevant tactical work is considered to include specification, supplier selection and agreements. When an order uses an existing supplier agreement, supplier selection and contracting do not need to be repeated for that order. Logistics and time recorded under EXC (the historical PO's rework and aftercare code) are excluded from the Measure analysis.

Both Arno's and Dennis's work will be recorded and preserved. The resulting workload profile will cover and measure Arno's operational purchasing work, but will not capture his entire workload.

4. Method

Research Design/Approach

The study combines qualitative process investigation with quantitative profiling and evaluation of one selected intervention. *Define, Measure, Analyse, Improve and Control* (DMAIC) provides the overarching structure, following de Mast and Lokkerbol (2012). *Define* and *Measure* establish the relevant purchasing process and Arno's workload profile. *Analyse* uses both buyers' evidence to investigate contributing factors and select a bounded improvement component. Within *Improve*, *Design Science Research Methodology* (DSRM) guides the objectives, design, development, demonstration and evaluation of the proposed AI-supported prototype, following Peffers et al. (2007). *Control* translates findings into recommendations for use and monitoring.

AI could support information extraction, document comparison, validation, or routine decisions. The technical form, level of autonomy, and integration requirements will be determined by the specific aim and the actual workflow in the observed case.

Sources of data and data collection

Source	Data and collection	Purpose
Observation on Arno's work (operational buyer)	Structured observation of tasks, active time, volume, interruptions, and decisions; case questions	Establish his included workload profile

Observation on Dennis's work (tactical buyer)	Targeted observation and case walkthroughs, recording tasks, inputs, outputs, decisions, and exchanges with Arno	Identify relevant task patterns, handoffs, and possible causes
Purchasing documents from Hytech-Pommec	Review SOPs, forms, and formal process documents	Compare documented and observed practice
Purchasing records and systems at Hytech-Pommec	Relevant requests, POs, confirmations, and available Exact or Orbis information, subject to access	Trace supported connections and assess feasibility
Evaluation cases	Record Arno's handling, review, correction, and outcomes; record changes to affected tasks in Dennis's work separately	Evaluate the primary outcome and effects on other roles

Structured continuous observation follows time-and-motion principles, including explicit coverage and task transitions (Zheng et al., 2011).

5. Data analysis approach(es)

Workload profile and process analysis

For sub-question 3, active handling time is the primary quantitative indicator in the exploratory profile of Arno's included purchasing work. Frequency and volume complement this indicator; judgement, uncertainty, and expertise notes inform difficulty. Interruptions describe workflow conditions. Where existing notes provide support, repeated work will be considered qualitatively within the same scope. Time alone does not represent the full workload construct. Raw records remain unchanged. Derived data excludes logistics, EXC, and unreliable durations. Timing checks address start/end evidence and overlap; classification confidence is separate. Included minutes are summarized by family, activity, and purchasing stage. Time shares are calculated from included minutes; occurrence rates require verified observation exposure. Unresolved and combined episodes retain time without duplication. Dennis's evidence remains separate because unequal coverage prevents comparisons of the overall workload. Process maps, cases, and follow-up questions will investigate the workflow for sub-question 1 and factors contributing to workload for sub-question 3, and inform selection for sub-question 4. Analysis will trace information and decisions through both roles, examining missing inputs, clarification, and overlapping responsibilities where supported. Proposed causes require supporting evidence. The selected component may involve Dennis, while Arno's workload remains the primary outcome.

Selection and requirements of the improvement

For sub-question 4, candidates must meet conditions for permission, data access, reference outcomes, risk control, and study time. The analysis will assess whether AI has a justified role. Candidates will be compared on the share of Arno's workload that the improvement could reduce, business relevance, data readiness, AI suitability, evaluation feasibility, and the need for human expertise. This can include effects on Arno's subsequent work within the validated purchasing boundary; EXC and logistics remain excluded.

Criteria, score anchors, and weights will be agreed upon with the academic supervisor before ranking; sensitivity analysis will assess uncertain evidence and plausible weights. Johan will validate business feasibility. Exact and Orbis checks will establish access and interface requirements. For sub-question 2, inputs, outputs, implementation conditions, review controls, and responsibilities will be specified with affected buyers, including Dennis, where relevant.

Evaluation of workload and quality

For sub-question 5, the improvement will be compared with the current practice using justified workload indicators and quality criteria. Before development and formal testing, the evaluation will specify cases, units, comparators, improvement thresholds, and quality acceptance rules. If selected, handling time will include preparation, review, correction, and all other attributable active work within scope. Equivalent incoming cases remain eligible if the intervention prevents Arno from working.

Development and evaluation cases remain separate. Matched cases and balanced condition order will be used where feasible; sample size and analysis depend on available evidence. Results will report workload indicators, uncertainty, and quality. Relevant AI costs will be scoped after selection, distinguishing estimates from measured costs. Effects on Dennis will be reported separately. The company will agree to acceptable transfers before evaluation; transferred work will not be presented as department-wide savings. Success requires predefined workload improvement while maintaining quality relative to current practice; no benefit remains a valid finding

6. Deliverables

Anticipated practical implications for the company

The project will help Hytech-Pommec understand which purchasing tasks contribute to Arno's workload and how relevant tasks and information exchanges involving Dennis affect that work. This understanding will support the selection of a single purchasing task or a coherent process step for AI-supported redesign.

For the selected component, the project will deliver a proposed process design, an AI-supported prototype, and an evaluation against current practice. The evaluation will assess its effects on Arno's workload and purchasing quality, including effects on Dennis's work where relevant. Based on these findings, the company will receive recommendations on whether and under what conditions the proposed solution could be used or further developed. The intended practical benefit is a feasible way to reduce Arno's workload while maintaining purchasing quality.

Anticipated theoretical insights

The study aims to provide context-specific evidence about how AI-supported changes to purchasing tasks and information exchanges affect an operational buyer's workload and purchasing quality. It will examine information requirements, verification, and human review in the selected component. Relating the findings to Van Weele's stages will help explain how the intervention operates across relevant purchasing roles. The discussion will address the limits of the single-company setting, observation coverage, and included work.

References

Reference list in APA format

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Part B – planning

The BEP comprises 420 study hours for 1BEPIEX. Figure 1 shows the Gantt chart, which is organized according to DMAIC.

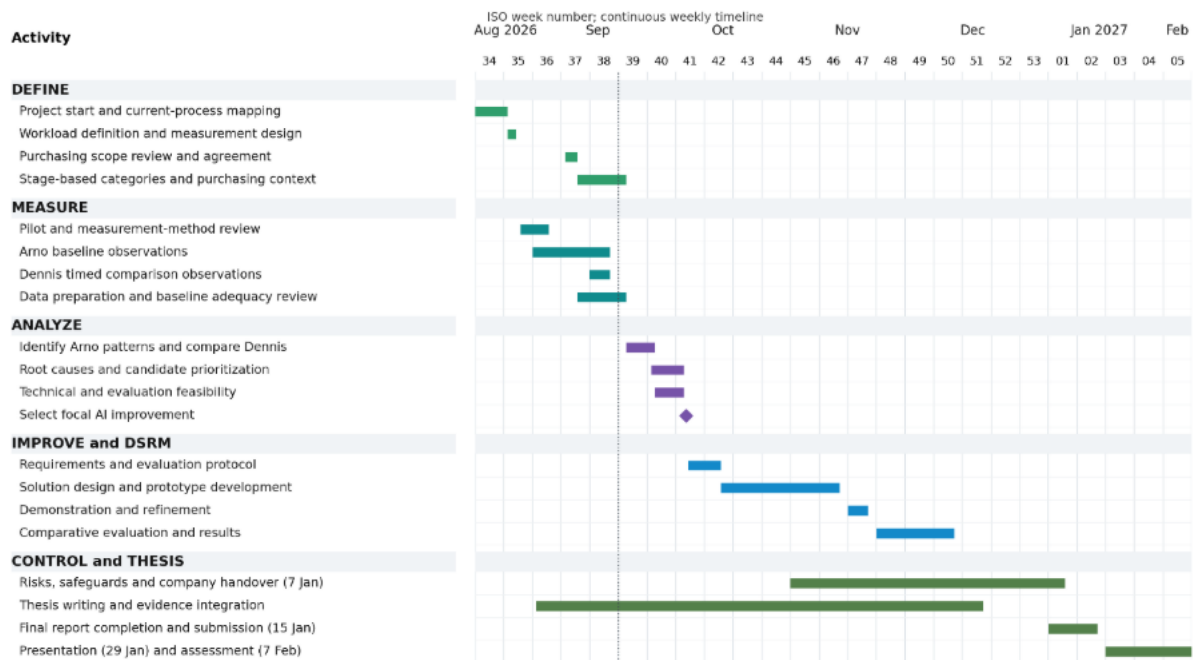


Figure 1: GANTT chart.

Part C – reflection at start BEP

1. Planning and Organizing

Experiences: In the Multi-Disciplinary CBL dashboard project (4CBLW00, 2024-2025), I worked on interface design, evaluation and reporting. My self-study reports recorded time for meetings, design changes and writing, which helped make my contribution visible. However, arranging evaluations depended on other people. SSA 13 records that, after other groups did not respond to my emails, I contacted fellow students to arrange participation. This experience showed me that a task list needs to include dependencies, follow-up moments and a workable alternative when access is delayed.

Feedback: In my earlier P&PD reflection, I identified prompt responses and deadline reminders as useful contributions to the team. My recorded peer feedback also asked for more active participation and more detailed input. That feedback suggests that being available and reminding others about deadlines is only part of organizing shared work. I also need to communicate what is ready, what is uncertain and which decision is needed next. The project records give me concrete examples to review, although they do not establish that I consistently planned well.

Self-assessment: I rate myself 6/10. I can organize tasks and keep records, but my planning is still fairly basic. I need to improve estimates, prioritize essential work and anticipate dependencies. In this BEP, access to buyers, usable observations and supervisor decisions will influence progress. A detailed schedule will only help if I update it when these conditions change.

Learning goals: During this semester, I want to maintain a realistic weekly plan that connects tasks to the next research milestone. I aim to identify access or review dependencies at least one week ahead and make the effect of unfinished work visible before it threatens a deadline.

Action plan: Each Monday I will select three priority outputs, estimate their hours and identify any required input. On Friday I will compare planned and actual effort, record reasons for differences and update the next week. I will include a short progress and dependency summary in supervisor updates.

2. Writing

Experiences: During the CBL dashboard project, I contributed to the report introduction and the explanation of the design process. This gave me practice translating design work into a written explanation. Looking back at the self-study reports, I can follow what I did, but some descriptions need clearer grammar, more precise wording and a stronger connection between the problem, evidence and design decision.

Feedback: My earlier P&PD reflection states that my writing conveyed the main message, while fuller explanations were needed to make it understandable to others. The student dashboard evaluation also identified unclear labels and explanations, including the meaning of the progress bar and peer recommendations. Those comments concerned the interface, but they reinforced a writing lesson: a term that is familiar to me can still be unclear to its reader. I need to explain meaning rather than assume shared understanding.

Self-assessment: I rate myself 5.5/10. I can produce a useful first draft, but I need deliberate revision to make it clear and academically precise. I sometimes describe activities without sufficiently explaining why they matter. For the BEP, I particularly need to distinguish observations, interpretations and conclusions, and avoid implying that a proposed benefit has already been demonstrated.

Learning goals: My goal is to write paragraphs with one clear point, supporting evidence and an explanation of its relevance. Before each major draft review, I want consistent terminology for

workload, handling time and purchasing quality, and a clear connection between the research questions and methods.

Action plan: I will reserve a weekly writing block and draft sections alongside the research. For each section, I will outline the argument before writing and check terminology, grammar and source support afterwards. At draft reviews, I will ask whether the reader can identify the main claim and understand its evidence. I will keep a revision log of recurring problems and compare an early methods section with the revised version in November and the final report in January.

3. Presenting

Experiences: In the CBL project, I prepared part of the final presentation about the development of the prescriptive dashboard. The saved condensed script covers slides 2 to 9 in approximately three minutes and describes several design changes and their rationale. Preparing it required me to select what an audience needed to know. Reviewing it now, I see how easily a short presentation can become crowded when it tries to cover too many changes, sources and claimed benefits.

Feedback: In my earlier course, I received full marks for the presentation-skills component and no detailed feedback. It also identifies confidence and composure as areas I wanted to improve. I regard the recorded result as encouraging, but it gives limited information about particular aspects of delivery. For this BEP, I therefore need more specific feedback on pace, structure, clarity and answers to questions. I should also verify every numerical result before including it in a slide or script.

Self-assessment: I rate myself 6/10. I can prepare a structured explanation and supporting slides. My main development needs are selecting the most relevant information, speaking with greater composure and explaining uncertainty clearly. A prepared script helps me organize my thoughts, but I want to become more comfortable answering questions about the reasoning behind a result.

Learning goals: By the final presentation, I want to explain the purchasing problem, method, main result and limitation within the agreed time. A listener unfamiliar with the project should be able to repeat the main conclusion and understand the evidence supporting it.

Action plan: From October, I will practice a three-minute project update every two weeks and record selected rehearsals. I will ask a peer or supervisor which message they remember and where the explanation became unclear. I will check timing and reduce unnecessary detail after each rehearsal. Before the final presentation, I will complete two full rehearsals and practise questions about case selection, quality criteria and uncertainty. I will compare feedback from early and final rehearsals to assess progress.

4. Collaborating

Experiences: The CBL dashboard project required me to coordinate design work with several teammates. Such as designing and prototyping with Maciej and sharing code and styling approaches during pair programming and also discussions with Veerle and Maud about stakeholder feedback, as well as writing contributions that supported their report sections. These experiences showed me that completing my own task is not sufficient when another person needs a clear explanation or a usable handover. Decisions about what to change also needed agreement within the team.

Feedback: My earlier reflection in CBL indicate to contribute more actively in meetings and provide more detailed input. It also reports improvement in participation in group chats and in sharing completed deliverables. I recognise progress here, but the earlier feedback remains relevant.

Responding to messages is easier than raising a concern, explaining an alternative or checking that everyone has understood a decision. I need to take a more active role in these discussions.

Self-assessment: I rate myself 6/10. I am able to work with others and respond to feedback, but I still need to make my reasoning and progress clearer. During the BEP, I will work independently while depending on buyers and supervisors for access, interpretation and decisions. I need to respect their time and communicate questions in a form they can answer.

Learning goals: I want to make a relevant contribution in each substantive project meeting and leave with a shared understanding of actions and responsibilities. I also want to raise disagreements or uncertainties early enough for others to respond before they affect the research.

Action plan: Before each meeting, I will prepare a short update, a clear question and, where appropriate, proposed next step. Afterwards, I will record decisions, action owners and deadlines and check important interpretations with the relevant person. Once a month, I will ask a supervisor or collaborator for one example of effective communication and one improvement. I will review these examples alongside meeting notes in November and January to assess whether my participation and handovers have become clearer.

5. Dealing with Scientific Information

Experiences: In the CBL dashboard project, I helped investigate scientific literature on prescriptive learning dashboards. The shared research-paper overview identifies contributions by Veerle and me and includes my notes on potential benefits and limitations. This gave me experience connecting literature to a design problem and considering differences between users. However, a collection of article summaries does not automatically form a justified argument. Reviewing the project documents also reminds me to check whether broad statements about design benefits are supported by the specific study being cited.

Feedback: The stakeholder meeting notes record openness to exploring an LLM, alongside scepticism about how prompts would be generated and a request to discuss privacy with the LLM group. This feedback concerned the proposed application and highlighted questions that needed investigation. It reminded me that an appealing technical idea still requires evidence about how it works and whether it fits the setting. In the BEP, I need to turn such questions into focused searches and explicit design or evaluation requirements.

Self-assessment: I rate myself 6/10. I can find relevant papers and extract useful ideas, but my critical appraisal and synthesis need improvement. I need to examine methods, samples, outcome definitions and limitations more consistently. Findings from a student dashboard cannot simply be transferred to procurement. Similarly, a reduction in handling time must be assessed together with the quality of the purchasing outcome.

Learning goals: My goal is to build a traceable literature argument for workload measurement, process analysis and the selected AI intervention. For every central claim, I want to identify the supporting source, explain why it is relevant and state any limitation that affects its use.

Action plan: During the next six weeks, I will critically read at least two relevant papers per week. I will maintain a literature matrix with the question, method, setting, findings, limitations and relevance to my study, and record search terms and selection reasons. Before using a reference, I will check the original text and bibliographic details. At methods and draft reviews, I will ask the supervisor to challenge key source-to-claim links. Progress will be evaluated through corrected claims and a more coherent synthesis, rather than the number of citations alone.

Part D - Declaratio of scientific conduct



Declaration concerning the TU/e Code of Scientific Conduct for the Bachelor's final project

I have read the TU/e Code of Scientific Conduct¹.

I hereby declare that my Bachelor's final project has been carried out in accordance with the rules of the TU/e Code of Scientific Conduct

Date

22/09/2026

Name

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Signature

Submit the signed declaration to the student administration of your department.

¹ See: <http://www.tue.nl/en/university/about-the-university/integrity/scientific-integrity/>

The Netherlands Code of Conduct for Academic Practice of the VSNU can be found here also.
More information about scientific integrity is published on the websites of TU/e and VSNU